DESIGN EXCEPTIONS GUIDELINE

Technical Standards Branch
Alberta Transportation and Economic Corridors



September 2026

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Design Exceptions Guideline

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List of Changes

The following page is reserved for documenting changes to this version of the Design Exceptions Guideline.

Document Revision	Description	Date
1	Add Need Assessment of Design Exception and Revise Design Exception Request Form	September 2018
2	Update Need Assessment of Design Exception, update Design Exception Request Form	September 2026

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ACKNOWLEDGEMENTS

The following were members of the Design Exception Task Force. The task force was instrumental in developing and modifying the Alberta Transportation and Economic Corridors (TEC)Design Exception process in 2016/2017.

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DEFINITION AND BACKGROUND

Design Exceptions (DE) are defined as instances where a designer has chosen or is requested to use a parameter, guideline, principle or product which is different from the currently published standards and/or practices. DEs can be initiated at any stage of a project, and may be initiated by the Consultant or by the Department. DEs accepted at an early stage may be revisited and re-submitted at a later stage if conditions change or new pertinent information becomes available.

The purpose of DEs is to allow for deviations from normal design standards or practices to be made in a thoughtful and consistent way where warranted by the project specific conditions and constraints. This practice allows for innovation, “flexible design” and/or “context sensitive design” to be applied to Alberta highways in a way that gives appropriate consideration to highway safety, risks and mitigation. By following a consistent documented process the Department is aware of common deviations from normal practices and is able to undertake timely reviews of any practices as warranted. This would normally include a review of any effects from the implementation of DEs in the past. The use of a DE process provides a good rationale to support design decisions, which may be useful in the future if the deviation from normal practice is questioned. If there emerges an ongoing recurrence of similar types of Design Exceptions being received, the Department may consider the need to change their standards.

Consultants are encouraged to suggest innovative designs and/or value optimizing adaptations to designs. Examples where flexibility in design has been accepted in the past include:

- Reduced traffic exposure/risk on low volume roads;
- An unconventional barrier layout for intersections near hazards;
- Reduced design speed;
- An unconventional layout to accommodate log haul vehicles at an intersection;
- Median acceleration lanes;
- Stopping sight distance on directional ramps;
- Steep gradients in rolling or mountainous terrain;
- Intersection sight distances and/or stopping sight distances on existing paved roads, etc.

A listing of previously submitted Design Exception summaries is available on the Department’s website at: https://www.alberta.ca/design-exceptions

Key references for this section include:

- Mitigation Strategies for Design Exceptions, Aug 2007, published by U.S. Department of Transportation, Federal Highway Administration.
- Geometric Design Guide for Canadian Roads, June 2017, published by Transportation Association of Canada.

Concepts, processes and criteria from both of these references have been adopted for use by TEC and are reproduced in this Guide

Design Exceptions Triggered by the Department

If the Department triggers a DE, this should be clearly recorded with supporting documentation provided by the Department. The Department may trigger a DE due to a constraint in the budget, schedule, or possibly because the Department is aware of an unconventional solution that should be considered for the project. Therefore, the Department may request unconventional options to be assessed, and/or an option not recommended by the Consultant. The Department should identify contemplated DEs in the Terms of Reference for projects where possible. However, in some cases the need for a DE may not become apparent until later in the process. Some examples of Department-initiated DEs include: using a single lane bridge, deferring construction on an interchange, limiting pavement structure thickness, undertaking a trial project, installing a trial product, allowing overlay of narrow pavement, etc.

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KEY DESIGN ELEMENTS FOR DESIGN EXCEPTIONS

It is not practical to evaluate every design detail as a DE. Consequently, TEC has chosen 16 principal criteria to be evaluated as they have significance for the operation and safety of a highway. The department may identify additional criteria on particular projects as warranted by the circumstances. The principal criteria are as follows:

- Design Speed
- Lane Width
- Shoulder Width
- Bridge Width
- Horizontal Alignment
- Superelevation
- Vertical Alignment
- Grade
- Stopping Sight Distance
- Intersection Sight Distance
- Cross Slope
- Vertical Clearance
- Lateral Offset to unprotected hazard
- Design Vehicle
- Vulnerable Road User Facilities (width and continuity of trails and sidewalks)
- Access Control

NEED ASSESSMENT OF DESIGN EXCEPTION

DE is not automatically required for every incidence of deviation from normal design standards or practices. The Department's **P**roject **S**ponsor should inform the Subject Matter Experts (SME) in the Department for any proposed deviation from the current standards and guidelines. Based on the initial assessment, SMEs will indicate whether or not a "formal" DE request is required. A list of examples where DE may not be required includes the following items:

- Steeper backslope outside of clear zone due to right of way constraint, requiring no barrier.
- Uses of 90 degree short radius guardrail due to other alternatives being not feasible.
- Pilot projects (for example: Edgeline Rumble Strips) accepted by the department.
- Usage of trial products accepted by the department through the Products List process.
- Use of 3R/4R sag and crest K values on existing paved roads (instead of new construction values) with acceptable collision historical and/or operational experience.
- For bridge projects, design exceptions need not be completed if the appropriate background and support information is included in the Bridge Conceptual Design Report or the Structure Alternatives Report, and the report is reviewed and accepted by the appropriate SME.

If DE is not required, the Department's Project Sponsor should document the relevant information for future reference. The documentation should include an appropriate level of engineering analysis, evaluation of alternatives, risk assessment, consideration of mitigation strategies and final course of action. The file can be retained with the Project Sponsor and a copy should be provided to the appropriate SMEs. SMEs may require a DE if the documentation is considered worthwhile for a particular project or to support future enhancements to technical standards or practices. Similarly, the Consultant may trigger the need for a DE due to a desire to obtain a more formal acceptance of the deviation from the Department. If a DE is **required**, the Department's Project Sponsor and the Consultant should follow this **Design Exception Guideline** to prepare and submit the formal documentation.

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DESIGN EXCEPTION PROCESS

The process to evaluate and document a DE should be thorough and consistent. Figure 1 provides the outline of a good process showing six basic activities, which are explained in the subsections below.

Figure 1 - Design Exception Process



Calculate Estimated Costs and Impacts of Normal Design Standards

A designer or planner should begin by assessing the implications of undertaking a project while meeting normal standards and practices for all of the key design elements. This will provide a benchmark for comparison of any other alternatives that are proposed. The comparison may include agency costs such as the timing and extent of capital costs, operating and rehabilitation costs as well as implications on highway safety, the environment, the local community and the highway network. It is acknowledged that it may not be practical or appropriate to assess all aspects of all Design Exceptions; however, the following should generally be considered and documented:

- Road safety impacts
- Road user costs (including vehicle operating costs, travel time costs etc.)
- Construction costs including right-of-way and utilities
- Deferred rehabilitation costs (the magnitude and timing)
- Sensitivity to context and community values
- Preservation of historical or cultural resources
- Natural environment impacts
- Consistency with the subject highway and the larger network

Develop, Evaluate and Compare Alternatives

It is normal practice to develop a number of alternatives to help in evaluating the implications of adopting a different value for one or more of the key design elements. This will be useful in determining which changes are favourable to achieve the desired effect while minimizing or eliminating any undesirable impacts. This activity, when fully documented, can form the “rationale” to support the decision to accept a DE and therefore is a fundamental component of any DE request. Ideally the AT Benefit Cost Analysis User Guide and Model should be used to evaluate the merits of each alternative and assist in formulating a recommended deviation from standards with minimal adverse effects. If the AT Guide and Model are not practical or necessary as agreed by the Project Sponsor for a particular DE, then the economic argument can be presented in an alternative way however the same economic principles and values should be used. Road safety risks and predicted effects are considered a significant outcome however they should be assessed together with all other outcomes.

Risk Assessment

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It is recognised that the inability to provide the usual design standards or follow the usual practices may increase the risk to operations and traffic safety. This is partly because of the possibility of surprising the road user but also because the customary factors of safety built in to the design criteria may be reduced. Designers should assess any proposed DE to gain a better understanding of the risks including the exposure, likelihood and severity. The following aspects should be included in a comprehensive risk assessment:

- Traffic Volumes – higher volumes lead to greater exposure and higher probability of safety incidents and road user costs in general.
- Traffic Composition – the percentage of all types of vehicles should be considered as well as the presence of any special vehicles such as log haul trucks, Long Combination Vehicles and oversized vehicles and loads.
- Posted Speed – lower posted speeds generally reduce the risk, provided there is good compliance.
- Severity of DEs - small deviations are less significant than large deviations.
- Length of the DE – the effects can differ depending on the length over which the DE is applied.
- Multiple DEs – There can be a compounding effect if two or more DEs are applied together.
- Duration of DE – a short-term duration such as a construction season will result in much less traffic exposure than a permanent DE.
- Other Risk Factors – other geometric or environmental factors may affect the severity of the risk.
- Existing Safety – What is the existing safety record over the previous 5 to 10 years?
- Predicted Safety – What is the predicted collision type, frequency and severity for each alternative based on the best available current models calibrated for the location?

Some of the key design criteria are more critical than others and it is generally found that many elements are less critical at lower speeds such as 70 km/h or less. The following observations have been made regarding design criteria in general:

- Vertical grade is not a very critical criterion in the urban environment especially if there is lighting and somewhat lower speeds e.g. 70 km/h or less.
- Vertical curves, especially sag curves, are often found to be less critical.
- Lateral offset to obstruction is often adequately addressed by provision of the standard shoulder width rather than the lateral offset required to obtain full stopping sight distance, provided that suitable roadside barrier protection is used.

Evaluate Mitigation Measures

On projects that include one or more DEs, designers will have a good understanding of the potential risks to safety and operations. Based on this understanding, designers are required to evaluate appropriate mitigation measures to address the risks. Mitigation may include provision of warning, improvement of other features to compensate for the reduction in a particular feature, modification of certain roadside features to lessen the severity of any adverse results or combinations of some or all of the above.

TEC has accepted a series of documents under the umbrella of Methods of Reducing Collisions on Alberta Roads (MORCOAR) which can be found on the Department's website. Many of the strategies listed under MORCOAR are for general application to highway, road and street design, however, especially in the case of retrofits, many of these strategies may also be used as mitigation measures for DEs. The MORCOAR Phase 2 Final Report contains listings of collision reduction measures for various roadway elements, common traffic management issues, common collision types and vulnerable road users. The suggested context for each type of measure is shown as well as examples of how they can be applied, typical economic returns on the investment and supporting references.

Table 1 shows some possible mitigation measures for each of the 16 key design criteria. Designers may employ other mitigation measures that are not listed in the table.

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Table 1 - Key Design Element Mitigation Measures

Design Element	Objective	Potential Mitigation
Design Speed	Reduce 85th percentile speed to design speed.	Gateway treatments; cross-section elements modified
Lane Width and Shoulder Width	Optimize safety and operations by distributing available cross-section width.	Select optimal combination of lane and shoulder width based on site characteristics
	Provide advance warning of lane width reduction.	Signing
	Improve ability to stay within lane.	Recessed pavement markings; centreline rumble strips; painted edgeline rumble strips; recessed lane delineators (cat's eyes)
	Improve ability to recover if vehicle leaves lane.	Shoulder rumble strips; safety edge on pavement
	Reduced crash severity if vehicle leaves the road surface.	Remove-relocate fixed objects; shield fixed objects & steep slopes; use traversable slopes; use crash-worthy or break-away devices on roadside hardware
	Provide space for enforcement or disabled vehicles.	Pull-off areas
Bridge Width	Provide advance warning and delineation of narrow bridge. Improve visibility of narrow bridge, bridge rail and lane lines.	Signing; object markers; reflectors on approach guardrail and bridge rail; post-mounted delineators; high-visibility bridge rail; bridge lighting; enhanced pavement markings
	Maintain pavement on bridge that will provide safe driving conditions.	Skid-resistant pavement; anti-icing systems
	Reduce crash severity if vehicle leaves the roadway.	Crashworthy bridge rail and approach barriers
	Provide space for disabled vehicles or emergencies on long bridges.	Pull-off areas in special cases
	Provides quick response to disabled vehicles or emergencies on long bridges.	Real-time traffic monitoring
Horizontal Alignment and Superelevation	Provide advance warning.	Signing; dynamic curve warning systems; pavement marking messages
	Provide delineation.	Chevrons; reflectors on barrier; post-mounted delineators
	Improve ability to stay within the lane.	Widen the roadway; lighting; skid-resistant pavement; enhanced pavement markings; centreline/shoulder rumble strips; painted edgeline rumble strips
	Improve ability to recover if vehicle leaves the lane.	Shoulder rumble strips; flatter than usual pavement edge and/or sideslopes
	Reduce crash severity if vehicle leaves the roadway.	Remove or relocate fixed objects; traversable slopes; breakaway safety hardware; shield fixed objects and steep slopes
Vertical Alignment and Grade	Provide advance warning.	Signing
	Improve safety and operation for vehicles ascending/descending steep grades.	Climbing lanes; downgrade lanes
	Capture out-of-control vehicles descending steep grades.	Escape lanes
	Improve ability to stay within lane.	Enhance pavement markings; delineators; centreline rumble strips; shoulder rumble strips; painted edgeline rumble strips
	Improve ability to recover if vehicle leaves lane.	Shoulder rumble strips
	Reduce crash severity if vehicle leaves the roadway.	Remove or relocate fixed objects; traversable slopes; breakaway safety hardware; shield fixed objects and steep slopes
	Address drainage on flat grades.	Provide special gradient on ditches; adjust gutter profile on curbed cross-sections
Stopping Sight Distance and Intersection Sight Distance	Mitigate sight distance restrictions.	Signing and speed advisory plaques (crest vertical curves); lighting (sag vertical curves and intersections); cross-section elements to manage speed; adjust placement of lane within the cross-section (horizontal restriction)
	Improve ability to avoid crashes.	Wide shoulders; wider clear recovery area
	Improve driver awareness on approach to intersections.	Advance warning signs; rumble strips in lane on approach to stop or yield condition; dynamic warning signs; larger or additional STOP/YIELD signs; intersection lighting; pavement marking message such as "STOP AHEAD" etc., as warranted
Cross Slope	Provide warning of slick pavement.	Signing
	Improve surface friction.	Chip seal
	Improve drainage.	Chip seal
Vertical Clearance	Advance warning.	Signing
	Reduce impacts with low structures.	Over-height detection systems with flashing lights; alternate routes; large vehicle restrictions
Lateral Offset to Unprotected Hazard	Improve visibility of objects near the roadway.	Delineate objects; lighting
	Improve visibility of the lane lines and shoulder lines.	Enhanced pavement markings
Design Vehicle	Advise drivers of restricted use.	Signing; gateway treatments to restrict height and/or width; consult and agree with other road authorities as needed
Vulnerable Road User Facilities	Advise users of restricted width or discontinuity.	Signing; provide alternate routes
Access Control	Achieve the desirable spacing and geometrics of access for the function of roadway.	Reduce the element of surprise for road users through the use of Gateway Treatments, signing, delineators or other features in advance and at accesses

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Document, Review and Accept or Reject

Documentation should clearly describe the location and nature of the DE. This includes identification of the implications of building within normal standards and practices versus any other alternatives that are considered or recommended. Discussion should include functional aspects, unique context, consideration of alternative solutions, evaluation of risk and exposure to risk. The impacts and effects should be weighed against each other, preferably using analytical tools such as collision prediction models and the Department's Benefit Cost Analysis Model which can be found at: https://www.alberta.ca/benefit-cost-model-and-user-guide. The designer must be able to support the recommended action based on the analysis in the current context. It is critical that the documentation provides a rationale to support the recommendation and identifies appropriate measures to mitigate the potential risks associated with the DE.

AT uses a standardised documentation protocol and decision process (for acceptance or rejection) to provide consistency and repeatability in decisions. Table 2 shows the typical items that should be considered at a minimum. Other items and issues may need to be included depending on the nature of the DE and the context.

Requests may be submitted with the DE form available at https://www.alberta.ca/design-exceptions. The form is also shown in Figures 2a-b. It can be modified as required, but must be signed, stamped and validated by a professional engineer licensed with APEGA. Regardless of the format used for the submission, the important thing is that the issues referenced in the process are addressed by the party preparing the DE request.

It is prudent that the proponent of a DE has a dialogue with the appropriate person in TSB and the Project Sponsor in advance to ensure that the submission addresses pertinent issues that the department is aware of (in addition to issues that the consultant is aware of). This step is expected to result in a less onerous and more timely process overall

Note: DE requests may occur at the planning stage. Current information on planning practices in the Department is not readily available for some of the subject areas. Please contact the Executive Director responsible for Planning or the appropriate Regional Director where clarification is needed.

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Figure 2a - Example of Design Exception Request Form (1 of 2)

DESIGN EXCEPTION REQUEST FORM

Date:
Project:
Region:
Project Sponsor:
Consultant:
NOTE: complete, modify and/or provide additional information as required.

Project Stage

( ) Functional Planning ( )Preliminary Design () Detailed Design ( )Construction
( ) Traffic Impact Assessment (development) ( ) Access Management ( )Other_____Please specify

Project Type

() Functional Planning ( )New Construction ( )Reconstruction ( )Paving/Surfacing ( )Bridge

( )Operations ( )Geotechnical ( )Environmental ( )Other_____Please specify

Project Data (typically required for all projects):

Project Description				
Highway No.			Control Section	
Km Posts	From:		To:	
Length of Project				
Chainage (if applicable)	From:		To:	
Chain Direction	South to North		West to East	
Design Designation				
Service Classification				
Basic or Existing Right-of-Way				
Existing Traffic Vol.	AADT		ASDT	
Projected Traffic Vol. (years)	AADT		ASDT	
Design Vehicles			Vehicle Distribution	
Design Speed			Posted Speed	

Example of information required for geometric design:

Cross-Section:	Existing Width		Width after Overlay	
	3R/4R (Suggested Min. Width)		Current Standards	
	Backslope		Sideslope	
	Ditch Width			
Level of Service:	Existing		Projected	
Horz. Alignment	Min. Radii		Proposed Radii	
Vert. Alignment	Min. K Crest Curve		Proposed K Crest Curve	
	Min. K Sag Curve		Proposed K Sag Curve	
	Max Grade		Proposed Grade	
Existing Passing/Climbing Lanes				

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Figure 2b - Example of Design Exception Request Form (2 of 2)

DESIGN EXCEPTION REQUEST FORM

Collision History (if applicable): (Period Year to Year)

() Segment ( )Interchange ( )Intersection ( )Bridge ( )Other_____Please specify

Collision Data	Non Animal	Animal	Total
Collison Rate			
Collision Frequency			
Collision Severity Breakdown	# Fatal	# Injury	# PDO
Other (Type): Please specify			

Other Criteria - Please Specify

Details and Supporting Documentation of Design Exception

Provide drawings, analysis, evaluations, cost estimates, rationale, justification, mitigation, etc. and supporting documentation as required.

Prepared by:

Professional Engineer Stamp Permit to practice Stamp

Accepted:

Executive Director of Technical

Standards Branch/Date

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Table 2 - Design Exception Documentation

Category	Details
Basic Information	• A description and details of the project including the type of project (functional planning, new construction, 3R/4R projects, bridge, pavement surfacing, etc.), the location of the project, lengths and limits of the project including the km posts, highway service class, design speed, posted speed, cross-sections, and other improvements to be considered. • Site plans, profiles, sketches, detailed drawings, and/or photographs of the DE and the alternatives considered. • Current and future projected traffic volumes, growth rate, traffic composition, and Turning Movement Diagrams (if applicable).
Design Element(s) and Criteria	• Describe the proposed or planned work(s) requiring a DE, stating the design element(s) to which the DE applies. • The minimum value or range and source of that value. • The proposed value and the degree to which the standard is being modified.
Explanation	• Reasons for the DE • Site constraints (if applicable). • A cost estimate to build to standard versus to the DE, and include cost estimates of alternatives. Use tables, charts and drawings as needed to illustrate impacts. Depending on the economic impact of the proposal on the amount or timing of capital, maintenance, road user or other costs, an economic analysis may be warranted. This is not a mandatory requirement but rather is dependent on the project and the nature of the DE. In many cases a simplified benefit cost analysis may be appropriate. In all cases the principles used should be consistent with the department's Benefit Cost Guide even though use of the Guide's spreadsheet is not mandatory. • A summary of the current standards/practices that are not being followed and what alternatives were considered and evaluated. If the DE has been initiated by the Department, the Department shall provide the supporting rationale. If the Consultant is aware of a better than standard option while doing the assessment or conceptual design, presenting that option (at a high level – minimal detail) is part of the basic expectations of delivery from a skilled, professional partner.
Impacts	• The potential impacts, if any, the exception may have on other standards or practices (e.g. safety and traffic operations). • Implications to future planned improvements to the roadway or corridor that may need to be considered. • A detailed review of the collision history within the project limits. Address and summarize the safety and operational implication and/or collision experience related to the proposed work(s) for the DE. Quantify the substantive safety of the location and how it compares to similar facilities. • Assessment of the exposure and risk with respect to traffic volume, location, severity (predicated and worst case scenario), duration, etc. The evaluation may involve an assessment beyond the project limits. The evaluation may also include a Road Safety Audit (RSA). Normally if an RSA has been completed on a project, this would be submitted together with the DE request. In the event that an RSA has not been done and the Executive Director of Technical Standards Branch (TSB) feels it is needed to support the request, an RSA may be required. The performance of an RSA at the DE stage is not the usual practice and can be expected to delay the process beyond the usual response time of three weeks.
Mitigation	• Mitigation measures that were considered (e.g. safety enhancements such as signing, markings, barriers, etc.). • Recommended mitigation measures and any additional operational or maintenance costs associated with them. Include drawings as appropriate. • The effect that mitigation measures are expected to have to address the risks. • Practices implemented in other jurisdictions may be recommended as a potential solution if warranted by the conditions in a particular design.
Supporting Documentation	• Identify any research or other technical resources that were consulted as part of the evaluation process.

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Monitor and Evaluate In-Service Performance

Performance of DE locations should be monitored to gauge the effectiveness of mitigation measures and to add to the knowledge about the safety and operational effects of DEs and mitigation measures. Pending a favourable outcome, design standards and practices may be fine-tuned as a result of extensive in-service experience with DEs. Monitoring may be done as part of a special safety assessment program or undertaken as a component of regular highway operations management.

RECOMMENDATION AND ACCEPTANCE

DEs on AT projects are reviewed at a project level and regional office prior to being submitted to Technical Standards Branch for a final review and decision (either to accept or reject). The purpose of this process is to ensure independence as well as to provide communication back to Technical Standards Branch about pressures to deviate from normal standards and practices.

If a DE request has been prepared and advanced by a consultant to the Region (or other Project Sponsor), the supporting documentation shall be stamped by the appropriate professional in advance of submission to the Department. Depending on the nature of the DE, an appropriate professional may include areas of other practicing disciplines (besides engineering) from other professional associations, societies and/or organizations recognized in the Province of Alberta.

If a DE has been accepted at the planning stage, or through Geometric Assessment, Safety Assessment or Road Safety Audit, the need to revisit that decision at the design stage is to be determined by the Department and would be stated in the Terms of Reference. Planning decisions often warrant a re-visit for various reasons such as: the time elapsed since the planning work was undertaken, evolution of standards and practices, changes in adjacent development and urbanization etc.

In some cases, the Department may request a consultant to prepare a DE. The Department shall provide time constraints and financial constraints if they exist. Examples of those cases will be DEs triggered by funding limitations, unresolved utility issues between the Department and third parties, inadequate right-of-way resulting from unsuccessful expropriation or regulatory requirements. The Consultant has the right to refuse the work if they are uncomfortable with the Department's request.

All requests for DEs must be submitted by the Project Sponsor and the appropriate Regional Director or Executive Director to the Director of Roadway Engineering for acceptance by the Executive Director of TSB. The Executive Director of TSB may forward the DE request to SMEs including Planning (if applicable) to provide an independent review for recommending acceptance or rejection. All recommendations for acceptance by the Project Sponsor, appropriate Executive Director and/or SME(s) shall be submitted and documented by email or cover letter to the Director of Road Geometric Design prior to final acceptance by the Executive Director of TSB.

If the DE is accepted, the acceptance shall be signed off on the DE form by a Professional Engineer representing AT. When the DE is accepted, the Department considers it to be the standard for that element of that particular project.

The Department understands that there may be a change in the risk level when Consultants are working outside of the normal standards and practices. Consequently, the Department is willing to evaluate each DE on a case by case basis. If the DE is accepted, the Province will expressly agree to the deviation from normal standards/practices for that particular instance on that particular project.

The acceptance of all DEs shall be documented for future reference.

Appendix A is an example of a completed TEC DE request form for a particular project. This illustrates many of the elements described in Table 2. Values shown in the example were using standards of the time it was completed. New DE of similar nature will need to use values of present day standards.

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APPEAL PROCESS (BETWEEN CONSULTANT AND DEPARTMENT)

If the Consultant is in disagreement with the DE as requested by the Department, they may appeal to the Department in a process as follows:

- Communicate to the Project Sponsor (typically the Region) the rationale/justification for the disagreement with the DE.
- The Project Sponsor shall set up an appeal meeting with the Consultant and representatives from TSB and/or Planning, whichever is applicable. The purpose of the meeting will be to discuss the concerns of all parties involved, and to establish a joint solution that is generally agreeable to all, subject to fiscal and time constraints. The meeting shall be recorded and the minutes shall be provided as supporting documentation to facilitate the acceptance process of the DE. If an agreement cannot be reached in the appeal meeting, the decision may be elevated to the Executive Director of TSB. Alternately, the Consultant may choose not to proceed with the project, or the Department may choose to remove the work in question from the overall scope of the project; in both cases another party will have to be solicited to perform the unfinished work.
- If an agreement is reached, proceed with the solution established at the appeal meeting, providing all required documentation as per the usual DE request process (including the minutes from the appeal meeting).

TIMEFRAME FOR RESPONSE

If the DE application is fully documented (including a rationale, drawings, risk analysis etc. as applicable), the normal timeframe for response from TSB is three to four weeks. In the interest of getting a timely response, it is prudent for the Consultant or Project Sponsor to engage in initial discussions with TSB on the concept and clarification of required information prior to the formal submission.

DISPUTE RESOLUTION (INTERNAL TO THE DEPARTMENT)

In the event that an agreement cannot be reached between the Executive Director, TSB, and the sponsoring Executive Director on the DE, then it may be elevated to the Assistant Deputy Minister (ADM) of Planning Grants Engineering Services (PGES) for a final decision.

All requests must be fully documented (including the decision of the Executive Director, TSB) and submitted by the sponsoring Executive Director to the ADM of PGES. The Executive Director, TSB should be copied on the request.

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APPENDIX A EXAMPLE OF DESIGN EXCEPTION

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Alberta

DESIGN EXCEPTION REQUEST FORM

Date: September 19, 2025 (Revised March 30, 2026)
Project: Region: Peace
Project Sponsor: Rommel Directo, P.Eng.
Consultant: Egis Canada Ltd.

NOTE: complete, modify and/or provide additional information as required.

Project Stage

( )Functional Planning (**X**)Preliminary Design ( )Detailed Design ( )Construction
( )Traffic Impact Assessment (development) ( )Access Management ( )Other_____Please specify

Project Type

( )Functional Planning (**X**)New Construction ( )Reconstruction ( )Paving/Surfacing ( )Bridge
( )Operations ( )Geotechnical ( )Environmental ( )Other_____Please specify

Project Data (typically required for all projects):

Project Description	Highway 2 Realignment South of Dunvegan			
Highway No.	2		Control Section	68
Km Posts	From: 14.37		To: 18.73	
Length of Project	4.36 km			
Chainage (if applicable)	From:		To:	
Chain Direction	North to South			
Design Designation	RAU-212-110			
Service Classification	2			
Basic or Existing Right-of-Way	New Construction ROW TBD			
Existing Traffic Vol.	AA	2520	ASDT	
Projected Traffic Vol. (years)	AA DT	3235	ASDT	
Design Vehicles			Vehicle Distribution	
Design Speed	110 km/hr		Posted Speed	100 km/hr

Example of information required for geometric design:

Cross-Section:	Existing Width	13.6m and 17.3m (with 3.7m Climbing Lane)
	Design Guidelines Suggested Min. Width	12m
	Backslope	3.25:1
	Ditch Width	4m
Level of Service:	Existing	B
Horz. Alignment	Min. Radii	600m & 420m
Vert. Alignment	Min. K Crest Curve	250
	Min. K Sag Curve	55
	Max Grade	5%
Existing Passing/Climbing Lanes		1 x 3.7m wide

Classification: Public

Alberta

DESIGN EXCEPTION REQUEST FORM

Collision History (if applicable): (Period 2016 to 2020)
(X)Segment ( )Interchange ( )Intersection ( )Bridge ( )Other_____Please specify

Collision Data	Non Animal:	Animal:	Total:
Collison Rate	73.66 C/100M VKT	50.64 C/100M VKT	124.3 C/100M VKT
Collision Frequency	16	11	27
Collision Severity Breakdown	# Fatal: 0 # Injury: 4 # PDO: 12	# Fatal: 0 # Injury: 2 # PDO: 9	# Fatal: 0 # Injury: 6 # PDO: 21
Other (Type): Please specify			

Other Criteria - Please Specify

Details and Supporting Documentation of Design Exception

Classification: Public

1.0 EXECUTIVE SUMMARY

This document presents the design exception requests for the realignment of Highway 2:68 south of Dunvegan, Alberta. The project, commissioned by Alberta Transportation and Economic Corridors (TEC) and executed by Egis Canada Ltd., aims to address the geohazard risks associated with the current road alignment by shifting the alignment eastward. This realignment is necessary to mitigate the risk of landslides and ensure the safety and stability of the highway. The Functional Planning Study ("Highway 2 Realignment Study, July 2014, Stantec") identified that a design exception would be required. After reviewing the report, Egis agrees with Stantec's recommendation and as such has proceeded with this design exception request.

Two (2) Design Request approvals have been included within this document, including the following items:

1. Design Exception for a vertical profile outside of the recommended 5.0% for a RAU-212-110 Design Designation
2. Design Exception for a decreased design speed of 90 km/hr (posted speed limit of 80km/hr) through the project limits to increase safety due to a substandard curve.

Recommendation

The proposed realignment spans approximately 4.35 kilometers, beginning at kilometer 14.371 (just south of the bridge) and ending at 18.727 at the top of the hill. **The recommended design includes a maximum vertical grade of 6.9%,** which exceeds the specified guideline maximum of 5% by 1.9%. A design exception is requested to accommodate this design due to the steep topography of the project area and will help to balance slope stability, drainage, and excavation costs.

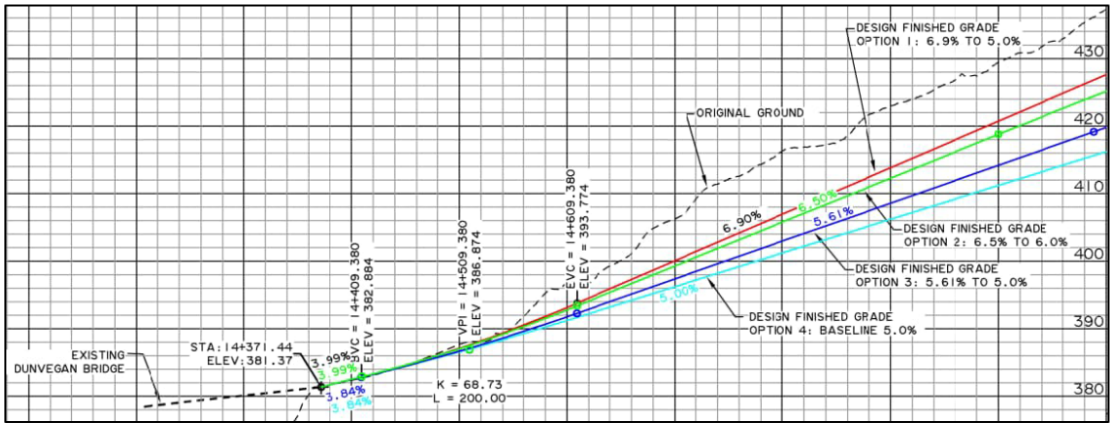


Figure 1. Vertical Profile Comparison

Analysis

Four design options were analyzed, each with varying vertical grades and corresponding cut volumes. Option 4, the baseline standard design, meets all specifications. The recommended option utilizes a 6.9% grade for approximately 1200 meters

Classification: Public

before transitioning to a 5.0% grade, which offers a balance between cost and safety. The total excavation cost for this option is estimated at **$62,505,000**. Due to excavation depths being 50+ meters deep, a slight adjustment to the vertical profile significantly impacts the overall excavation costs, as can be seen in table 1 below.

Table 1. Cost Comparison

Options	Cut Volume (m3)	Price ($/m3)	Total Cost
Option 1: 6.9% to 5.0%	**12,492,264**	**$5.00**	**$62,505,000**
Option 2: 6.5% to 6.0%	18,188,754	$5.00	$90,943,770
Option 3: 5.61% to 5.0%	34,183,283	$5.00	$170,916,415
Option 4: Baseline Standard Design (Maximum 5%)	44,873,792	$5.00	$224,368,960

In addition to cost, this document also addresses potential impacts and risks associated with the design exception, including traffic, safety operations and travel time.

Overall, the design exception for the Highway 2:68 realignment is a critical step in ensuring the long-term safety and functionality of this important transportation corridor. The proposed design balances the need for safety and cost-effectiveness, making it a viable solution for the challenges posed by the project area.

2.0 DESIGN EXCEPTION REQUEST 1

2.1 Basic Information

2.1.1 Project Details

Egis Canada Ltd. (Egis) has been retained by Alberta Transportation and Economic Corridors (TEC) to complete the Preliminary Engineering, Detailed Design, Tender Package Preparation, Contract Administration and Construction Supervision, Post Construction, and Warranty Inspection services for the Realignment of Highway 2:68 South of Dunvegan. This site is located near Dunvegan, Alberta and approximately 16 kilometers North of the town of Rycroft, Alberta. The shortest detour between Fairview and Rycroft, maintaining a service level 2 or higher roadway, is approximately 250 kilometers in length. Active geohazard slides (PH037, PH055, PH073) present a risk to the current road alignment and a realignment of the highway is required to mitigate this risk. The current realignment design is approximately 4.53 kilometers in length, with its limits beginning at kilometer posts 14.371 and ending at 18.727. Figure 2, shown to the right, is the alignment that has been selected by TEC for detailed design. Egis reviewed multiple design options, refer to the Options Analysis Report from May 2025 for more information on the selection process for each option.

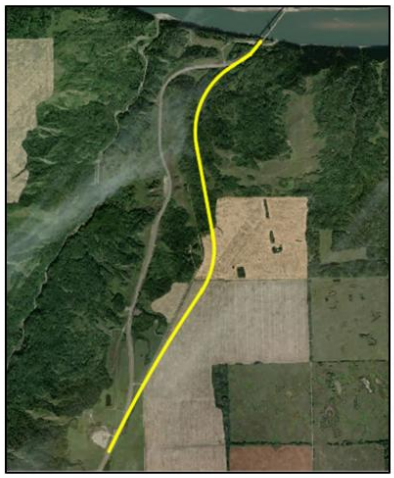


Figure 2. Horizontal Alignment

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2.1.2 Traffic Volumes and Service Class

Highway 2:68 serves as an important connector for communities north of the Peace River to larger communities in the south such as Rycroft and Grand Prairie. The Weighted Average Annual Daily Traffic (WAADT) volume for kilometer 11.992 to 20.061 of road is between 2520 units, 20.6% of the WAADT is comprised of heavy trucks. Using an existing WAADT of 2520 and a growth rate of 1.42%, sourced from the NESS Report of the area, a 20- year design WAADT of 3235 can be calculated. Future traffic volumes are expected to stay steady, increasing linearly. The service class for this control section is characterized as Level 2, with a level of service (LOS) of B.

2.1.3 Project Drawings

Figure 3 below shows the preliminary design cross section for the realignment of Highway 2:68. The complete plan and profile drawing set can be found at the end of this document.

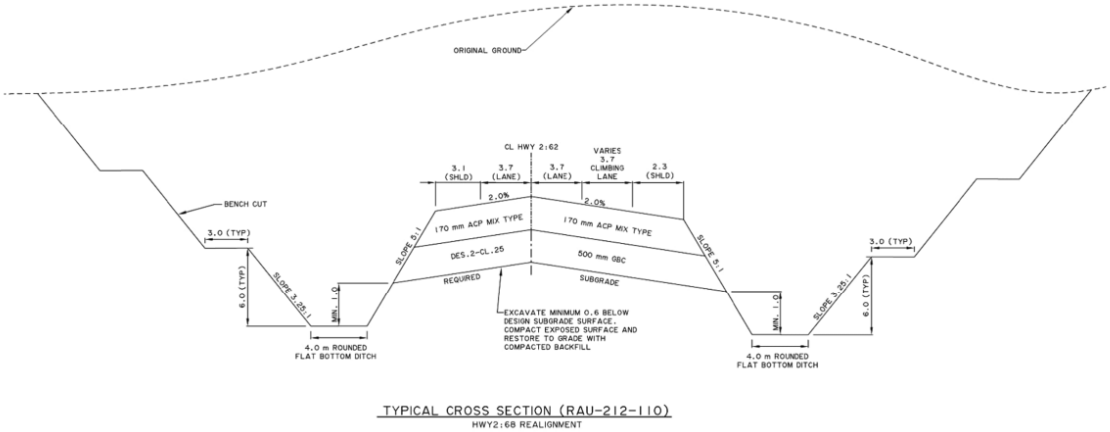


Figure 3. Design Cross Section

2.2 Design Exception Element(s) and Criteria

Due to the steep topography of the project area, it was expected that a design exception would have to be applied for to accommodate the vertical grade of the realignment. In the 2012 "Highway 2 Realignment Study South Dunvegan Valley" report by Stantec a maximum 6% grade was agreed to be utilized between Stantec and TEC. In the 2018 "PH037-1/PH073-1 Dunvegan South Landslides Preliminary Engineering Assessment" by Thurber a maximum 7% grade was utilized for the shorter 1.95km realignment. Egis' current realignment design has a maximum vertical grade of 6.9%, improving upon the maximum existing grade of 7.3% but still surpassing the specified guideline maximum of 5% by 1.9%. The specified maximum value was extrapolated from Table B-4-2a within the Alberta Transportation Highway Geometric Design Guide (HGDG). Egis

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is requesting a design exception to implement this maximum grade of 6.9%. The tables below provide a detailed summary of grade and stationing for each design that has been developed, including the existing road. These design options will be discussed more thoroughly in the next section of this document.

Table 2. Grade Comparison for Options

Re-Alignment			
Start (km)	End (km)	Grade (%)	Distance (km)
14.371	14.509	3.99	0.138
14.509	16.177	6.90	1.668
16.177	17.39	5.00	1.213
17.390	18.42	1.05	1.030
18.420	18.723	-0.02	0.303

OPTION 2 (6.5% MAX)			
Start (km)	End (km)	Grade (%)	Distance (km)
14.371	14.509	3.99	0.138
14.509	15.100	6.50	0.591
15.100	17.390	6.00	2.290
17.390	18.420	1.05	1.030
18.420	18.728	-0.02	0.308

OPTION 3 (5.61% MAX)			
Start (km)	End (km)	Grade (%)	Distance (km)
14.371	14.509	3.84	0.138
14.509	15.188	5.61	0.679
15.188	18.103	5.00	2.915
18.103	18.420	0.60	0.317

BASELINE (5% MAX)			
Start (km)	End (km)	Grade (%)	Distance (km)
14.371	14.509	3.84	0.138
14.509	18.225	5.00	3.716
18.225	18.420	0.49	0.195

Existing			
Start (km)	End (km)	Grade (%)	Distance (km)
14.371	14.519	1.80	0.148
14.519	15.071	6.50	0.552
15.071	15.526	3.70	0.455
15.526	16.404	6.90	0.878
16.404	16.674	0.40	0.270
16.674	16.825	3.80	0.151
16.825	17.071	2.10	0.246
17.071	17.341	4.90	0.270

2.3 Reason for Design Exception

2.3.1 Site Constraints

As mentioned previously within this report, there are geohazards (landslides) that exist within the project area that present a risk to the existing highway. Geohazard PH073 was not initially identified within the Stantec's realignment study, of which the proposed horizontal alignment is designed from. Egis has reviewed this slide location with Thurber Engineering to assess the risk PH073 poses to our current alignment, and have concluded that the geohazard is not in close enough proximity to affect the design. To mitigate the remaining hazards, Egis' proposed design will shift the road alignment to the east, however in doing so must traverse to the top of the Dunvegan Valley, resulting in a vertical grade that is steeper than the standard maximum. A defining factor within the design of this project is the cut material volume that will be generated. Millions of cubic meters of natural material will have to be excavated and graded in order to ensure slope stability and provide adequate

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drainage. The excavated material contributes to a significant proportion of the overall cost and any reduction or increase will have major implications. In the latest cost estimate, common excavation was 65% of the overall project cost.

2.3.1.1 Baseline Design

To compare each vertical profile, a baseline design was created that meets the design guide standards. This design has a maximum vertical grade of 5.0%. Table 3, shown below, provides quantification of this design if it were to be implemented.

Table 3. Baseline Design

Options	Cut Volume (m3)	Price	Total Cost
Baseline Design	44,873,792	$5.00	$224,368,960

Using an estimated price of $5.00 per cubic meter, this design would cost approximately $224,368,960 for grade construction.

2.3.2 Alternative Designs and Cost Estimate Comparison

Multiple vertical profiles have been developed and analyzed for the region within the preliminary engineering phase to reduce cost and quantities. After refinement, three designs consisting of a 6.9% slope transitioning to a 5.0% slope, a 6.5% to 6.0%, and a 5.61% to 5.0% were ultimately selected for further analysis. Reference the drawings in Appendix A for a visual representation. The horizontal alignment is the same for each option, therefore the vertical grade and corresponding cut volume quantities are the defining parameters of each. Table 4, shown below, presents the three mentioned options and the cut volumes associated with the designs.

Table 4. Design Options

Options	Cut Volume (m3)	Price	Total Cost
Option 1: 6.9% to 5.0%	**12,492,264**	**$5.00**	**$62,505,000**
Option 2: 6.5% to 6.0%	18,188,754	$5.00	$90,943,770
Option 3: 5.61% to 5.0%	34,183,283	$5.00	$170,916,415

Each of the options was designed with the intention of being a progressively higher threshold above the standard 5% maximum vertical grade. Investigating all three designs provides valuable information on the cost differences that would be incurred by relatively small changes to the vertical grade, as can be seen in Table 4.

Further analysis on the average speeds and time of journey for each option was also carried out. Using Figure B-5-3-3d from the Alberta Transportation HGDG, average speeds along specific points of the realignments were able to be interpolated. The results are shown in the graphs that make up Figure 4 on the next page.

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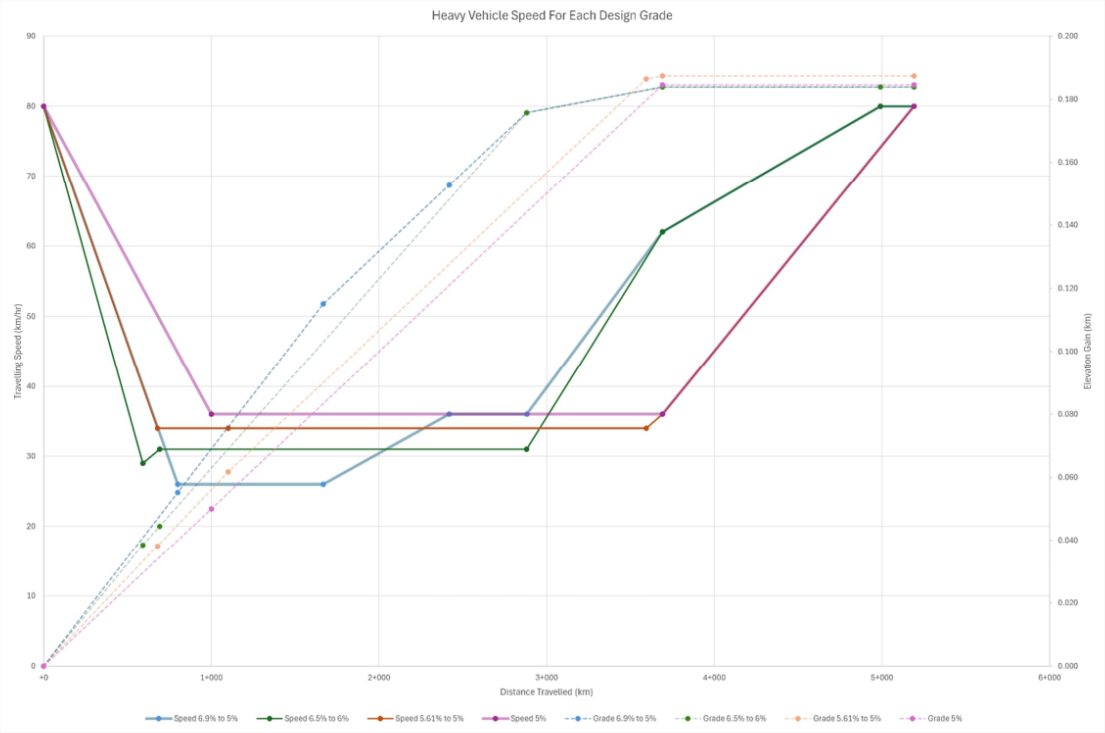


Figure 4. Heavy Vehicle Speed over each Design Grade

Figure 4 provides a visual representation of the average speed of a heavy truck (using an average mass/power ratio of 150 g/W (247 lb/hp)) that can be maintained through the site corresponding to each design option and continuing down the road until they regain an 80 km/hr travelling speed. Distance and elevation (height) are used as the parameters with the assumption that the vehicle begins at 80 km/h coming off the bridge. Each design vehicle experiences significant speed reductions as the major incline is encountered however, the 6.9% to 5% Option and the 6.5% to 6% Option can regain speed more rapidly around the 4+000 mark than others. Less speed differential between cars and heavy trucks is ideal.

Based on Figure 4, the total time for a vehicle to travel from km 14.371 to km 19.700 is shown below in Table 5. Note that Options 1 & 2 edge out Option 3 due to regaining speed faster as mentioned above. The baseline option of a steady 5.0% grade does have the shortest travel time overall, however due to the required excavation volume it has been included for information only. All new options have significantly shorter journey times relative to the existing road.

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Table 5. Comparison of heavy vehicle travel time across various grades

	Total Time (min)
Option 1 (6.9% MAX)	**7.31**
Option 2 (6.5% MAX)	7.33
Option 3 (5.61% MAX)	7.58
Baseline (5.0% MAX)	7.07
Existing Road	7.82

It should be noted that the existing vertical grade of Highway 2:68 within the work limits of this project is as high as 7.3%. The steepest vertical grade presented in Egis' design is 6.9%, which is an improvement on the current condition.

2.3.3 Cost Benefit Analysis

Alberta Transportation's Cost Benefit Analysis model was referred to while comparing each of the design grade lines. After review, the rehabilitation, maintenance, and road user cost benefit that would be associated with any of the options proposed is marginal. The grade difference between each option is not significant enough to present a notable difference regarding those parameters. While raising the vertical gradient does have the potential to increase collision costs even with mitigation measures that will be implemented (discussed in Section 5.0 of this report), these costs will be negligible when compared to savings from construction.

2.4 Impacts and Risks

2.4.1 Collision History

The collision history relevant to this project spans from kilometer 14.370 to 18.73 and covers the period from 2016 to 2020, during which a total of 27 collisions were recorded. Approximately 59% of these collisions were non-animal related, and 0 were fatal. The total collision rate is 124.3 collisions per 100 million vehicle kilometers traveled (C/100M VKT), while the rate for non-animal collisions is 73.66 C/100M VKT. The department's benchmark for the total rate is 82.2 C/100M VKT and the current rate is 45.5 above the benchmark rate. The benchmark collision rate for non-animal collisions is 32.2 and the current rate is 35.6 above the benchmark rate.

The most relevant collision types for this design exception are off-road, rear-end, struck object, and animal incidents.

Struck Object: This type is particularly pertinent for the steep grade when traveling northbound and approaching the bridge. Currently, there is a guardrail in the northbound lane (NBL). A review of the collisions involving striking an object indicates that none were due to the grade of the existing roadway. The primary issues were icy roads, non-standard road conditions due to construction, or debris on the roadway.

Off-Road: There was only one off-road collision in the northbound lane, where the driver was not focusing on the road and left the roadway unsafely. The rest of the collisions in the southbound lane were due to loss of control from icy road conditions.

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Rear-End: All 4 rear-end collisions were not due to current road profile and were caused by icy road conditions, non standard road conditions due to construction/parked vehicles on the shoulder and driver's not following roadway speed.

Animal: 39% of collisions are animal related. The general issue is with the position of the roadway and proximity to wildlife corridors and is not specific to this design exception and will be a consideration in the overall design of the project.

The most prevalent non-animal related issue causing collisions based on the history is icy conditions at this location. The impact of icy conditions will be reduced on the south side of the Peace River bridge due to the lower maximum grade and the new horizontal alignment having a less sharp horizontal curve before transitioning onto the bridge.

2.4.2 Potential Impacts

The implementation of this design exception will potentially impact the traffic and safety operations of this road if proper mitigation measures are not implemented (mitigation measures are discussed in the next section of this memo). There are two locations on the realignment that have exposure to risk, the boat launch access intersection and the entrance to the Peace River bridge. The boat launch access intersection will be a modified T-Intersection (Drawing IN-19647-01-P, shown at the end of this report), with road users travelling in the north bound (down grade) direction to potentially having to come to a complete stop if traffic in the opposite direction dictates it. Increasing the grade from a specified maximum of 5.0% to 6.9% creates concerns in this location. A vehicle travelling north bound will either have too much speed to be able to safely navigate the intersection, especially if unfamiliar with the route, or that a vehicle behind will not be able to stop in time if a vehicle in front has slowed down or come to a complete stop at the intersection entrance. The intersection sight distance (ISD) will be reviewed during the completion of the Preliminary and Detailed design reports to ensure that it conforms to the Chapter D requirements of the HGDG.

The entrance to the Peace River bridge poses a similar risk to the boat launch access intersection regarding speed. The tie-in to the bridge is at the conclusion of the 6.9% grade and there is a risk of brake failure, particularly in heavier vehicles, or the inability to stop in winter conditions. Due to the bridge only have 1 travel lane on each side of the centreline (8.5 meter total width), controlling vehicle speed will be critical. In order to assist with reducing vehicle speeds, the posted speed limit will be reduced from 100 km/h to 80 km/h in the approach to the bridge, with a horizontal curve radius of 400 meters.

2.4.3 Risk Assessment

A thorough comparison has been conducted between the proposed realignment and existing roadway to evaluate exposure, likelihood, and severity of risk. Key considerations include:

- **Traffic Volumes**
  Traffic volume is within expected volumes for this section of a Class 2 highway, and it is not expected to change due to the design exception or the project overall. Anticipated daily traffic near the boat launch remains low, which significantly reduces exposure and the probability of safety incidents associated with the steeper grade.
- **Traffic Composition**
  Low volumes also correlate with a limited presence of heavy or oversized vehicles. Although larger vehicles may still traverse the section, their reduced frequency helps mitigate associated risks. This highway section is divided into 78.3% Passenger, 2.1% Recreation, and 19.6% Commercial traffic.

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- **Posted Speed**
  The posted speed limit for the realigned section matches or is lower than the existing route. This improvement supports safer operations under the steeper grade conditions. The intersection will have a lower posted speed limit of 80km/hr which is deemed as the highest risk section with the steep grade.
- **Severity of Deviation**
  The proposed vertical grade exceeds standard guidelines; however, because this deviation applies only to a relatively short segment (1.6 km), and higher speeds are not facilitated, the overall safety impact is minimized.
- **Length of the Exception**
  The steep grade is confined to a concise realignment through the valley. This limited exposure duration provides less time for potential safety lapses, thereby reducing collision risk.
- **Exposure Time and Travel Efficiency**
  The new alignment shortens travel time through the corridor, diminishing the period during which vehicles are exposed to higher-risk roadway geometry, thus contributing to improved overall safety.

While adopting a steeper grade does introduce some risk—particularly for heavy or future high-volume traffic—the combination of low traffic volumes, controlled speeds, limited segment length, and reduced exposure time culminates in a net safety gain relative to the existing alignment.

2.4.4 Implications to future planned improvements

To Egis' knowledge there are no future planned improvements to this section of Highway.

Given the current age of the Peace River bridge, evaluating the construction of a replacement structure is warranted. Should a new bridge be built in the future, relocating the crossing to the east of the existing alignment may provide an opportunity to reduce the steep grade at the bottom of the south valley. Additionally, this adjustment could allow for a more optimal grade at the boat launch intersection.

2.5 Mitigation Measures

As previously mentioned within this memo, there are risks to safety and traffic operations present with the implementation of this design exception that can be mitigated.

Objective: Improve Safety and Operation on Steep Grades ascending/descending steep grade

The first mitigation measure implemented will be an auxiliary climbing lane in the southbound direction. This climbing lane will begin its taper shortly after the boat launch access intersection and conclude once the design vehicle has obtained a 80 km/hr merge speed. The exact start and end stationing of the climbing lane will be finalized within the detailed design report but are anticipated to be in the proximity of kilometres 14+530 to 18+150. Determination of the start and end locations of the Climbing Lane will be completed in accordance with Section B.5, Climbing and Passing Lanes, of the HGDG (Revised September 20. Heavier vehicles that will experience large speed reductions due to the steep grade will be able to utilize this lane without creating congestion in the standard travel lane. This will increase safety and reduce hazardous passing maneuvers due to vehicles being stuck behind slow moving heavy traffic.

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In the southbound direction at the intersection to the boat launch access, an auxiliary bypass lane will also be included. The intention behind this feature will be to allow vehicles turning left into the access to come to a complete stop safely in a delineated lane while road users continuing south can easily bypass them.

Objective: Provide Advanced Warning

Egis will investigate the currently non-functioning permanent variable message signs currently installed on elevated posts within the roadway ditches on either side of the river. If these signs are repaired or replaced before or during this project, they can be used to relay information about the steep south side road conditions, especially during icy conditions.

Additional signage can be installed along the northbound lane to warn the traveling public about the steep grade declining into the Peace River bridge. Advanced warning signage will include WA-21 "Hill" with WA-21B-T "Percentage Grade and Distance" for a 6.9% grade, and/or a WA-21A-T "Use Lower Gear".

Objective: Reduce crash severity if vehicle leaves the roadway

Throughout the curve and into the bridge transition, barriers will be installed to protect vehicles from leaving the road unsafely and/or striking objects outside of the roadway. It has been communicated by TSB that High Tension Cable Barriers (HTCB) are the preferred barrier system for this province. A feasibility study of potential barrier systems will be completed to ensure that the correct system is installed on the project, with preferred systems being High Tension Cable Barrier or Strong Post W-Beam systems. Recommended Barrier systems will be confirmed with Regional Construction staff prior to implementation on the project.

Implementing traversable 5:1 sideslopes will be used to reduce collision severity and promote potential recovery if a vehicle leaves the roadway.

Objective: Improve ability to stay within lane

The roadway will be painted with the region's preferred products for lasting visibility. Centreline and shoulder rumble strips and delineators will help keep drivers in their lane, especially during fog, mist, or smoke common in the river valley.

Implementing durable pavement markings at the boat launch intersection will enhance visibility for incoming vehicles. Clearly defined lane indicators will direct motorists towards the bypass lane, contributing to improved safety and a reduced likelihood of collisions at this location.

Objective: Capture out-of-control vehicles descending steep grades

A potential mitigation measure is the installation of a runaway lane at the bottom of the South valley before the bridge. This could address safety risks for large trucks slowing down due to the increased road grade. The department has previously examined the feasibility of runaway lanes with the current alignment. However, because of the curve when entering the bridge, a runaway lane near the transition to the bridge is not likely to be feasible without a major re-work of the existing topography given the proximity to the boat launch and the river. Table 6, shown below, summarizes the risks and mitigation measures.

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Table 6. Risk/Mitigation Matrix

Objective	Mitigation Measure / Action	Details
Improve Safety and Operation on Steep Grades	Auxiliary climbing lane (southbound)	Begins after boat launch access intersection, ends at top of hill; for heavier vehicles to avoid congestion
Improve Safety and Operation on Steep Grades	Auxiliary bypass lane (northbound at boat launch access)	Allows vehicles turning left to stop safely while others bypass
Improve Safety and Operation on Steep Grades	Design speed reduction	Approach to intersection reduced from 110 km/h to 90 km/h; radius of 400 meters does not meet 110 km/hr specification
Provide Advanced Warning	Variable message signs	Investigate/repair/replace signs above roadway; relay info about steep south side road conditions, especially during icy conditions
Provide Advanced Warning	Additional signage (northbound)	Warn about steep grade into Peace River bridge; WA-21 “Hill”, WA-21B-T “Percentage Grade and Distance” (6.9% grade), WA-21A-T “Use Lower Gear”
Reduce crash severity if vehicle leaves the roadway	Strong Post W-beam guardrail with delineators	Installed throughout curve and bridge transition
Reduce crash severity if vehicle leaves the roadway	Traversable 5:1 sideslopes	Reduces collision severity, promotes recovery if vehicle leaves roadway
Improve ability to stay within lane	Durable pavement markings	Region’s preferred products for visibility; centreline and shoulder rumble strips, delineators
Improve ability to stay within lane	Pavement markings at boat launch intersection	Enhance visibility for incoming vehicles; lane indicators direct motorists to bypass lane
Capture out-of-control vehicles descending steep grades	Runaway lane (potential)	Installation at bottom of South valley before bridge; feasibility limited by curve and proximity to boat launch/river

Classification: Public

2.6 Recommendation

The critical criteria in which our recommendation will be based on is the ability to optimize cost while providing a safe design. With the use of mitigation measures previously mentioned in this report, all the presented designs that require exceptions will produce a product that is safe for road users. However, Option 1 (maximum vertical grade of 6.9%) significantly reduces costs that would be associated with common excavation compared to the other options. It is approximately $161,863,960 less than the baseline design that meets the vertical grade design standards and $28,438,770 less than the next design (Option 2).

Egis recommends that Option 1 be carried forward as the preferred vertical design and is requesting this design exception approval to exceed the standard vertical grade by 1.9%.

3.0 DESIGN EXCEPTION REQUEST 2

Egis has reviewed the originally completed Stantec Planning Study, as well as discussing the project with TEC regarding the desirable posted speed limit descending into the Dunvegan valley. Based on our review, and in alignment with the originally completed and reviewed Stantec report, Egis requests that a design speed of 90km/hr through the majority of the project alignment be accepted due to a substandard curve approaching the bridge.

A copy of the originally prepared alignment options from Stantec (Figure 4.3, Highway 2 Realignment Study, South Dunvegan Valley Realignment Options) has been included within Appendix D for additional reference.

Based on our review and design, the geometry of the proposed re-alignment allows for a curve with a 400 metre radius immediately prior to the existing bridge structure tie-in. A 400m radius curve is only capable of meeting a maximum design speed of 90 km/h. The downhill approach grades entering the 400m metre curve are proposed to exceed 6%, making the transition from a 110 km/h design speed to a 90 km/h design speed considered to be unsafe over the project's steep decline. After reviewing the design, Egis is requesting a design exception to reduce the posted speed through the project corridor to a design speed limit of 90 km/h (posted 80 km/hr) beginning at Station 17+600.

Reducing the posted speed at the top of the valley further provides an advance indication of an abnormal roadway condition, including the approaching steep grade on the upcoming hill, and allows the posted speed transition from 100 km/h to 80 km/h to occur on a lower grade prior to descending into the river valley, increasing safety. A lower speed limit placed in conjunction with steep grade warning signage will further reinforce the complex situation about to be encountered by vehicles utilizing the highway.

Please accept this request for a reduced design speed of 90 km/hr with a posted speed limit of 80 km/hr through the indicated portion of the project.

Classification: Public

4.0 CONCLUSION & SIGNATURES

Egis is requesting approval of this design exception. Thank you for reading this document, should any questions arise please contact the undersigned.





March 30 2026

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Engineer in Training

Reviewed By:
Calvin Van Hill, P.L. (Eng.)
Manager, Design & Engineering

PERMIT TO PRACTICE
EGIS CANADA LTD.
RM SIGNATURE: _____
RM APEGA ID #: 63400
DATE: March 30 2026
PERMIT NUMBER: P013950
The Association of Professional Engineers and Geoscientists of Alberta (APEGA)

Accepted:

Paul.Buryn Digitally signed by Paul.Buryn Date: 2026.04.15 12:00:35 -06'00'

Date: _____

Executive Director of Technical
Standards Branch

Classification: Public